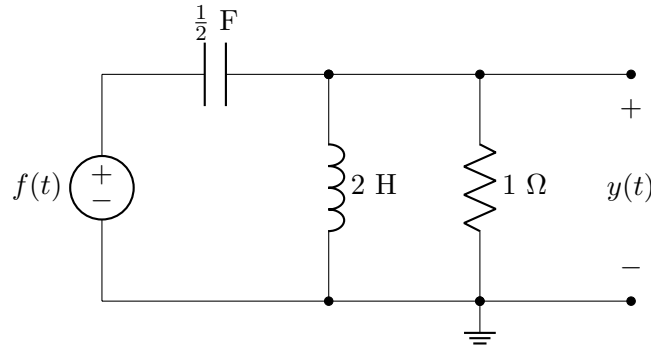


ECE 3210 quiz02

Name: _____ SOLUTIONS

1. (3 points) Consider the circuit below. Derive the ODE governing the system. Represent your answer in the form we used in class.



Solution: Start with node voltage analysis at the output node. We have

$$\frac{1}{2} \frac{d[y(t) - f(t)]}{dt} + \frac{1}{2} \int_{-\infty}^t y(\tau) d\tau + y(t) = 0.$$

We can differentiate both sides to get rid of the integral term and rearrange to get

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{df(t)}{dt}$$

2. (4 points) Derive the “zero-input” response of the system in the previous question assuming the initial conditions are $y_2(0^-) = 0$ and $y_2'(0^-) = 1$. Show all work.

Solution: Notice that $n = 2$, $b_2 = 1$, and $P(D) = D^2$. We can find the characteristic equation of the system by solving

$$\lambda^2 + 2\lambda + 1 = 0$$

which has a repeated root at $\lambda = -1$. Therefore, the zero-input response

$$y_2(t) = (C_1 + C_2 t)e^{-t}.$$

Applying the first initial condition we have

$$y_2(0^-) = C_1 = 0 \implies C_1 = 0.$$

Applying the second initial condition we need to take a derivative of $y_2(t)$ and evaluate at $t = 0^-$. We have

$$y_2'(t) = C_2 e^{-t} - C_2 t e^{-t} = C_2 (1 - t)e^{-t}.$$

Therefore, we have

$$y_2'(0^-) = C_2 = 1 \implies C_2 = 1.$$

Therefore, the zero-input response is

$$y_2(t) = t e^{-t}.$$

3. (3 points) Derive the impulse function $h(t)$ of the system in question 1. Show all work.

Solution: We need to put it all together. Using the equation

$$h(t) = b_2\delta(t) + P(D)y_2(t)u(t)$$

we have

$$h(t) = 1 \cdot \delta(t) + (D^2)y_2(t)u(t).$$

We can take the derivative of $y_2(t)$ to get

$$y_2'(t) = (1 - t)e^{-t}$$

and then take the derivative again to get

$$y_2''(t) = (t - 2)e^{-t}.$$

Therefore, we have

$$h(t) = \delta(t) + (t - 2)e^{-t}u(t).$$